Inventory Management System Project Report

Python, Java, and Data Analytics Implementation

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WEEK 6: June 8, 2026 – June 14, 2026

Project Overview

The Inventory Management System project focuses on designing and implementing an efficient software solution to manage stock levels, product tracking, forecasting, and reporting. The system integrates analytics tools to improve inventory decisions and operational efficiency.

Objectives

- Complete the frontend dashboard and fully integrate it with backend APIs.
- Implement analytics, reporting, and reorder suggestion features for inventory management.
- Enhance product search, filtering, and report export capabilities.
- Finalize backend validation, testing, and system readiness for demonstration.

Roles and Responsibilities

- Developed and integrated dashboard components for products, suppliers, transactions, and low-stock alerts.
- Implemented analytics, forecasting, reporting, and reorder suggestion functionalities.
- Improved backend validation, error handling, and maintained CRUD and authentication features.
- Conducted end-to-end testing, updated documentation, and finalized the project repository.

Tools and Technologies Used

Python, Java, Spring Boot, SQLite, SQL, HTML, CSS, JavaScript, Pandas, Matplotlib, Postman, GitHub, VS Code

Key Outcomes

- Successfully completed a demo-ready inventory management application with full frontend and backend integration.
- Implemented reporting, analytics, search, and reorder suggestion capabilities.
- Enhanced application reliability through comprehensive testing and validation.

Conclusion

Week 6 focused on completing the coding phase of the Inventory Management System and preparing it for demonstration. The finalized application integrated analytics, reporting, and inventory management features, resulting in a robust and user-friendly solution.

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